

City Smart Hospital Network Design

Computer Networks

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1 Team Contribution Statement

Each member contributed to a specific technical part of the hospital network design, configuration, verification, or documentation.

Student Name	Contribution	
Nada Mohamed	worked on	Designed the main hospital topology, including the placement of routers, switches, end devices, and the organization of the Administration, Medical, Outpatient, and Data Center zones. She also helped verify that all devices were connected according to the project requirements.
Ashar Salama	Worked on	Prepared the IP addressing scheme using FLSM /24 subnets for VLANs and point-to-point links. He also configured VLANs, trunk links, router sub-interfaces, and default gateways for the internal hospital departments.
Dolagy George	Worked on	Configured DHCP services on the internal routers for the connected VLANs. she also configured the Data Center DNS server and Web server, assigned static server IP addresses, and tested access to <code>www.cityhospital.local</code> .
Malak Osama	Worked on	Configured OSPF Area 0 routing between the hospital routers and integrated the Remote Radiology Clinic in Phase 2. She also performed routing verification and checked that the new clinic could communicate with the hospital network.
Nouran Hammam	Worked on	Configured NAT overload on <code>Core_Data_R</code> , the default route toward the ISP, and external connectivity testing. She also performed the failover test, captured traceroute and PDU screenshots, and organized the final report documentation

Table 1: Team member contributions.

2 Addressing Scheme & Justification

The hospital was assigned the private Class B address block `172.16.0.0/16`. According to the project requirements, the internal network was divided using Fixed Length Subnet Masking (FLSM), where each VLAN and each internal point-to-point router link receives a separate /24 subnet. This design keeps the addressing plan simple, readable, and easy to troubleshoot.

2.1 VLAN Addressing Scheme

Zone / Department	VLAN	Network Address	Default Gateway	Usable Hosts
Admin: Patient Records	10	172.16.10.0/24	172.16.10.1	254
Admin: Finance	20	172.16.20.0/24	172.16.20.1	254
Admin: HR	30	172.16.30.0/24	172.16.30.1	254
Medical: General Ward	40	172.16.40.0/24	172.16.40.1	254
Medical: ICU	50	172.16.50.0/24	172.16.50.1	254
Outpatient: Consultation Rooms	60	172.16.60.0/24	172.16.60.1	254
Outpatient: Guest Wi-Fi	70	172.16.70.0/24	172.16.70.1	254
Data Center: Servers	80	172.16.80.0/24	172.16.80.1	254
Remote Radiology Clinic	90	172.16.90.0/24	172.16.90.1	254

Table 2: VLAN addressing scheme using FLSM /24 subnets.

2.2 Point-to-Point Link Addressing

Router Link	Network	First Router IP	Second Router IP
Admin_R – Core_Data_R	172.16.100.0/24	172.16.100.1	172.16.100.2
Medical_R – Core_Data_R	172.16.101.0/24	172.16.101.1	172.16.101.2
Outpatient_R – Core_Data_R	172.16.102.0/24	172.16.102.1	172.16.102.2
Medical_R – Admin_R	172.16.103.0/24	172.16.103.1	172.16.103.2
Radiology_R – Admin_R	172.16.104.0/24	172.16.104.1	172.16.104.2
Core_Data_R – ISP_R	203.0.113.0/30	203.0.113.1	203.0.113.2

Table 3: Point-to-point and ISP link addressing.

2.3 Server Static IP Addressing

Server	VLAN	Static IP Address	Purpose
DNS Server	80	172.16.80.2	Resolves <code>www.cityhospital.local</code>
Web Server	80	172.16.80.3	Hosts the hospital intranet website
External Web Server	External LAN	8.8.8.8	Simulates <code>www.google.com</code>

Table 4: Static server IP address assignments.

2.4 Justification for Using /24 Subnets

A /24 subnet provides 254 usable host addresses. This is suitable for all required departments because the largest department, the General Ward, requires 120 devices, which is still below the 254-host capacity. Therefore, every department can fit inside one subnet while still leaving space for future growth.

The total number of internal /24 subnets used in the design is approximately 14: nine VLAN subnets and five internal point-to-point router-link subnets. Each /24 provides 254 usable addresses, so the total usable internal address capacity is:

$$14 \times 254 = 3556$$

The approximate number of actually needed internal addresses is 394. This includes the real department device counts, default gateway addresses, and point-to-point router interfaces. Therefore, the approximate number of unused addresses is:

$$3556 - 394 = 3162$$

This means that FLSM creates a large amount of unused address space. However, this is acceptable in this project because the organization owns a large private /16 block, and the main goal is to create a simple and consistent addressing plan.

The main advantage of FLSM is simplicity. Every subnet has the same mask, which makes configuration, troubleshooting, and documentation easier. It also makes the third octet easy to match with VLAN IDs. The main disadvantage is address waste, because smaller departments such as HR and the Data Center do not need 254 usable addresses. VLSM would reduce waste by assigning different subnet sizes, but it would make the addressing plan more complex.

3 Network Foundations

The hospital network is divided into multiple zones connected through a central data core. The internal hospital infrastructure, including the Administration Building, Medical Wards, Outpatient Clinic, Data Center, and Remote Radiology Clinic, represents the Local Area Network (LAN). These internal areas use private IP addressing from the 172.16.0.0/16 block.

The Wide Area Network (WAN) boundary begins at the connection between `Core_Data_R` and `ISP_R`. This link connects the private hospital network to the external ISP network. NAT overload is configured on `Core_Data_R` so internal private devices can access the external simulated internet server.

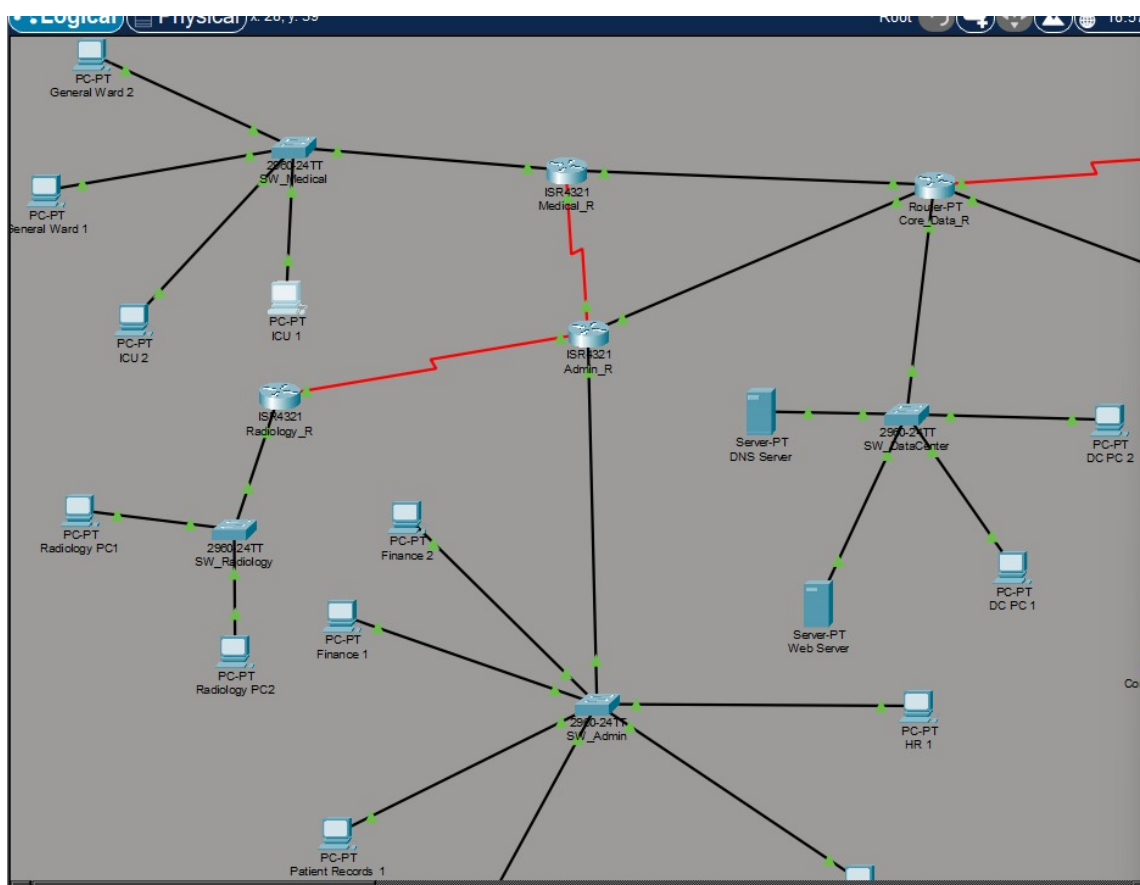


Figure 1: Logical topology overview of the hospital network.

VLANs were used to separate departments into different broadcast domains. For example, Finance, HR, Patient Records, ICU, and Guest Wi-Fi are placed in separate VLANs. This means that broadcast traffic generated in one department remains inside that department's VLAN instead of spreading across the whole hospital network. Router-on-a-Stick sub-interfaces were used on the zone routers to provide inter-VLAN routing while maintaining logical separation at Layer 2.

4 Medium Comparison

Zone 3 contains two different access methods: wired access for the Consultation Rooms and wireless access for the Guest Wi-Fi network.

The wired Consultation Rooms are connected through Ethernet switch ports assigned to VLAN 60. This provides stable bandwidth, lower latency, and more predictable performance. Wired access is more suitable for hospital staff and clinical devices because it is more controlled and less affected by interference.

The Guest Wi-Fi network is connected through a generic Packet Tracer Access Point attached to the outpatient switch. The access port is assigned to VLAN 70. Wireless access provides mobility and convenience for visitors, but it uses a shared medium. This means that performance may be affected by signal strength, interference, and the number of connected users. For this reason, Guest Wi-Fi is separated from clinical VLANs to improve security and prevent guest traffic from mixing with sensitive hospital operations.

5 Verification, Failover, & Design Reasoning

This section verifies that the network is operational and that the Phase 2 requirements were achieved. The verification focuses on internal connectivity, service reachability, the remote clinic expansion, and failover behavior after the primary Medical-to-Core link is removed.

5.1 Phase 2 Part A: Remote Clinic Expansion

The Remote Radiology Clinic was added as a new zone connected to **Admin_R**. A new unused /24 subnet, **172.16.90.0/24**, was assigned to the clinic. DHCP was configured for the clinic devices, and the new network was advertised in OSPF Area 0. This allowed the new clinic devices to communicate with the internal Data Center services.

The following screenshot demonstrates successful access to the hospital intranet website **www.cityhospital.local**. This confirms that DNS resolution, routing, and web service access are working inside the hospital network.

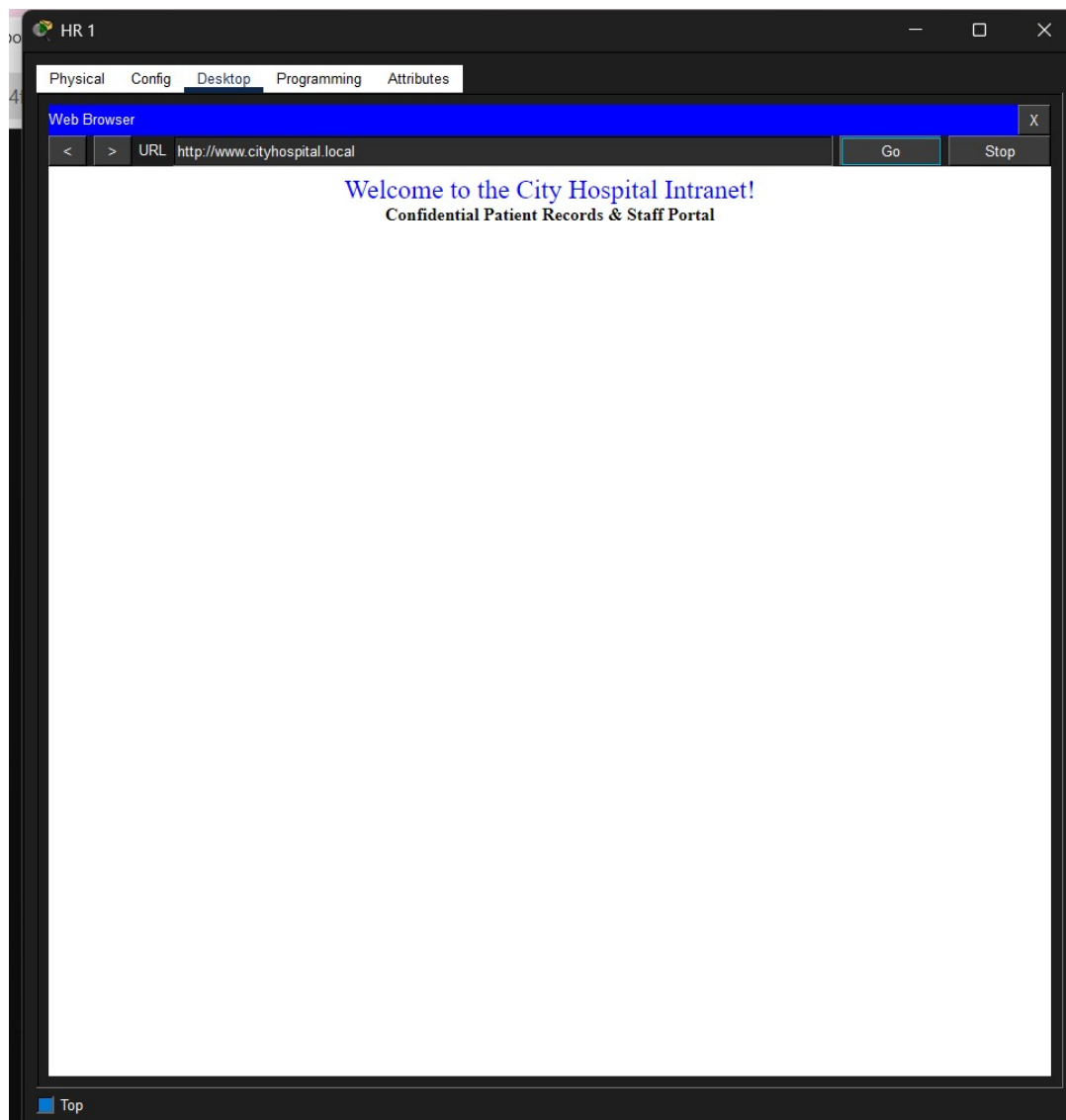


Figure 2: Successful access to the internal hospital intranet website.

5.2 Phase 2 Part B: Simulated Disaster and Failover

Under normal operation, traffic from the Medical zone to the Data Center follows the primary high-speed path between **Medical_R** and **Core_Data_R**. This is shown in the first traceroute result, where the route from ICU 1 to the Data Center web server uses:

$$172.16.50.1 \rightarrow 172.16.101.1 \rightarrow 172.16.80.3$$

After the primary link between **Medical_R** and **Core_Data_R** was removed, OSPF detected the topology change and recalculated the best available path. The second traceroute result shows that traffic changed to the backup route through **Admin_R**:

$$172.16.50.1 \rightarrow 172.16.103.1 \rightarrow 172.16.100.1 \rightarrow 172.16.80.3$$

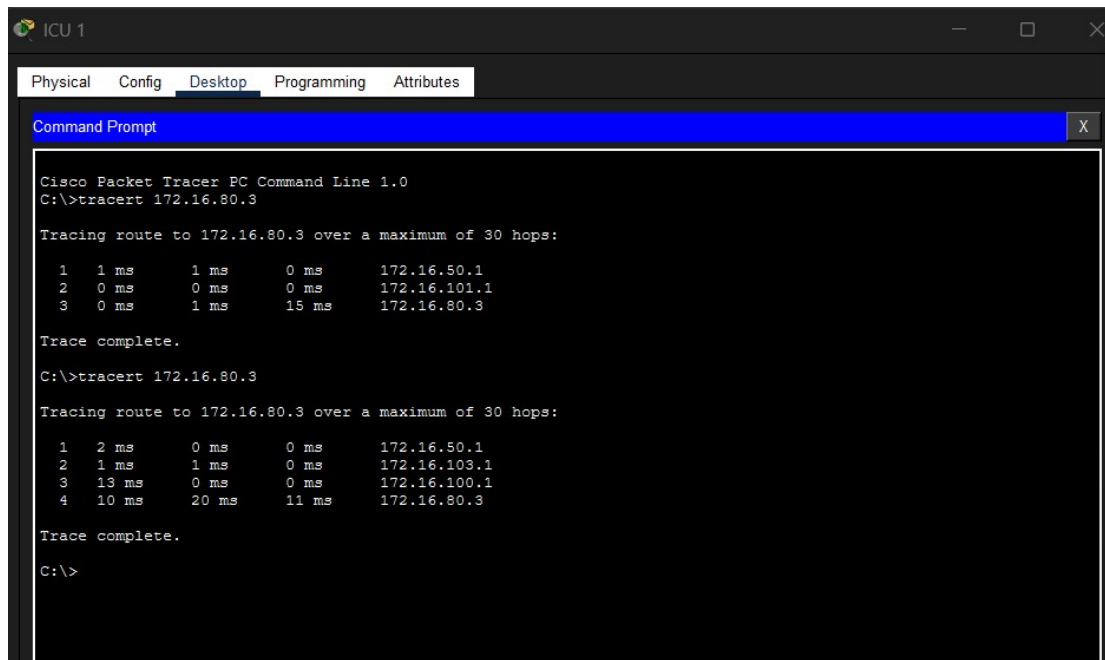


Figure 3: Traceroute from ICU 1 to the Data Center before and after failover.

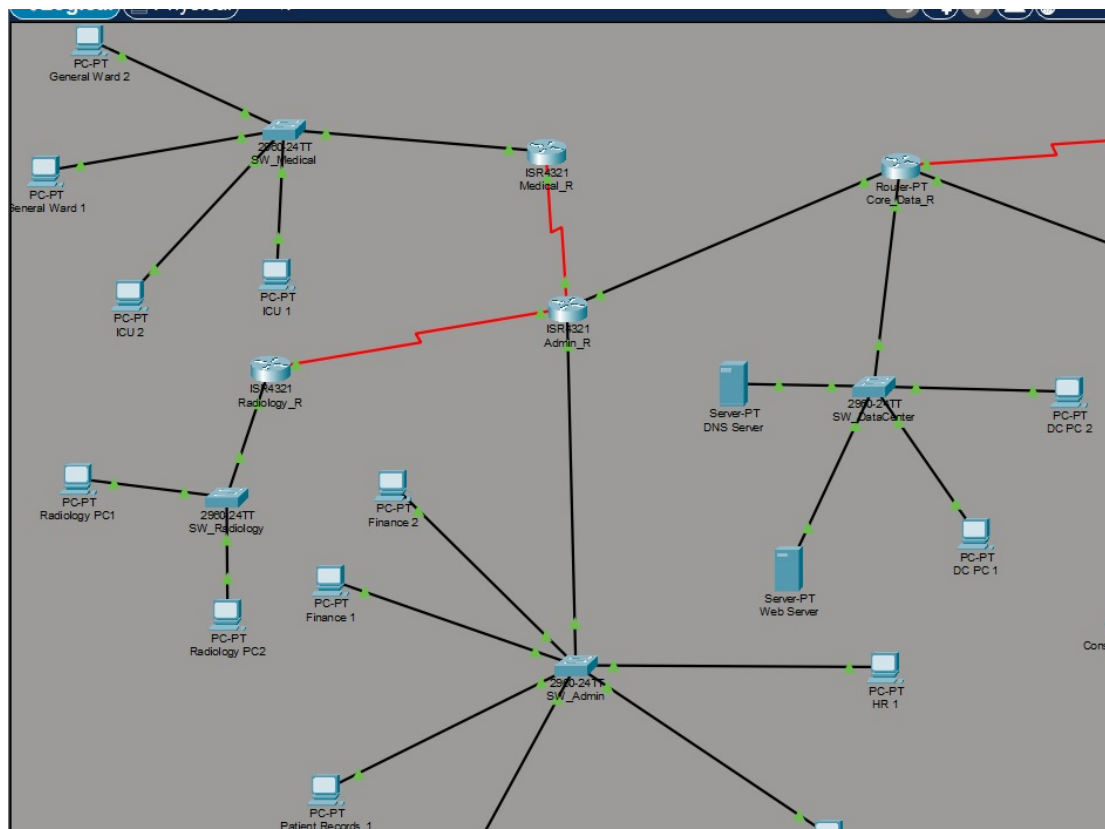


Figure 4: Topology after the primary Medical-to-Core link was removed.

The backup link was implemented between **Medical_R** and **Admin_R** instead of adding another direct link between **Medical_R** and **Core_Data_R** because it provides path diversity. If both links were connected directly to the same core path, the design would still

depend heavily on the same central connection area. By connecting Medical to Admin, the network creates an alternate path through a different router, improving resilience and demonstrating redundant routing.

OSPF reconverged after the failure because routers exchange link-state information and update their view of the topology. Once the primary Medical-to-Core link was removed, OSPF no longer considered that path reachable. It recalculated the shortest available route and selected the Medical-to-Admin-to-Core path. This is why the traceroute output shows different intermediate hops after the cable cut.

6 Protocol Analysis

Packet Tracer Simulation mode was used to capture traffic generated when an edge PC requested the hospital intranet page `www.cityhospital.local`. The capture showed DNS packets first, followed by HTTP traffic after the domain name was resolved.

6.1 DNS Request Analysis

The DNS capture shows a query for `www.cityhospital.local`. The PDU details show that DNS uses UDP as the transport-layer protocol, with destination port 53. UDP is suitable for DNS because DNS queries are usually small and require fast request-response behavior. Using UDP avoids the overhead of establishing a TCP connection before every name resolution request.

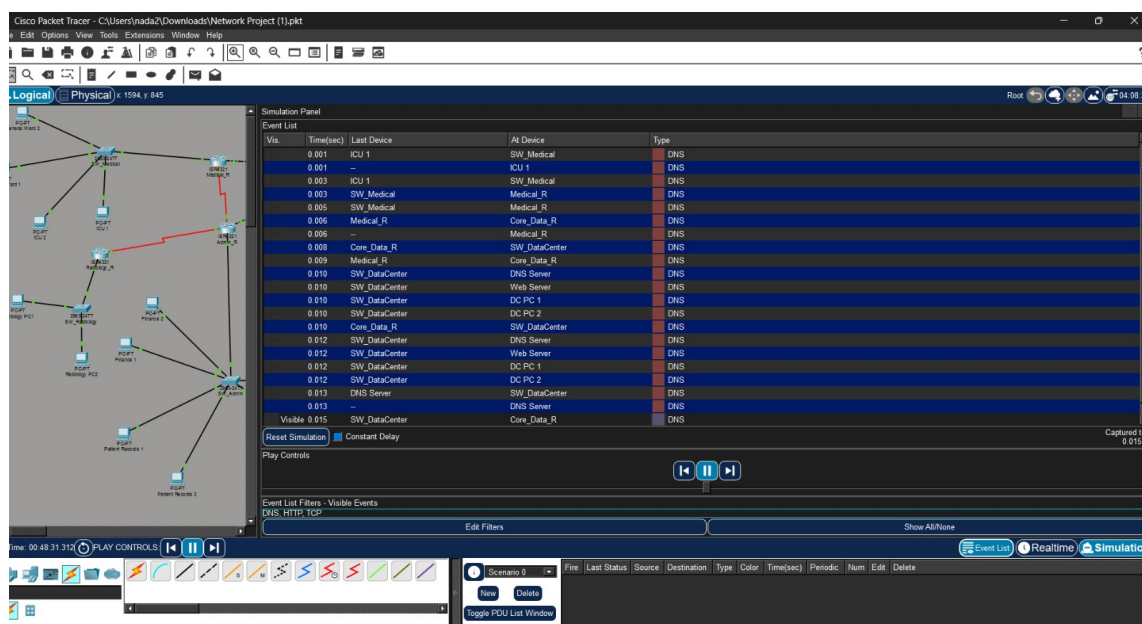


Figure 5: Packet Tracer Simulation mode showing DNS events during intranet access.

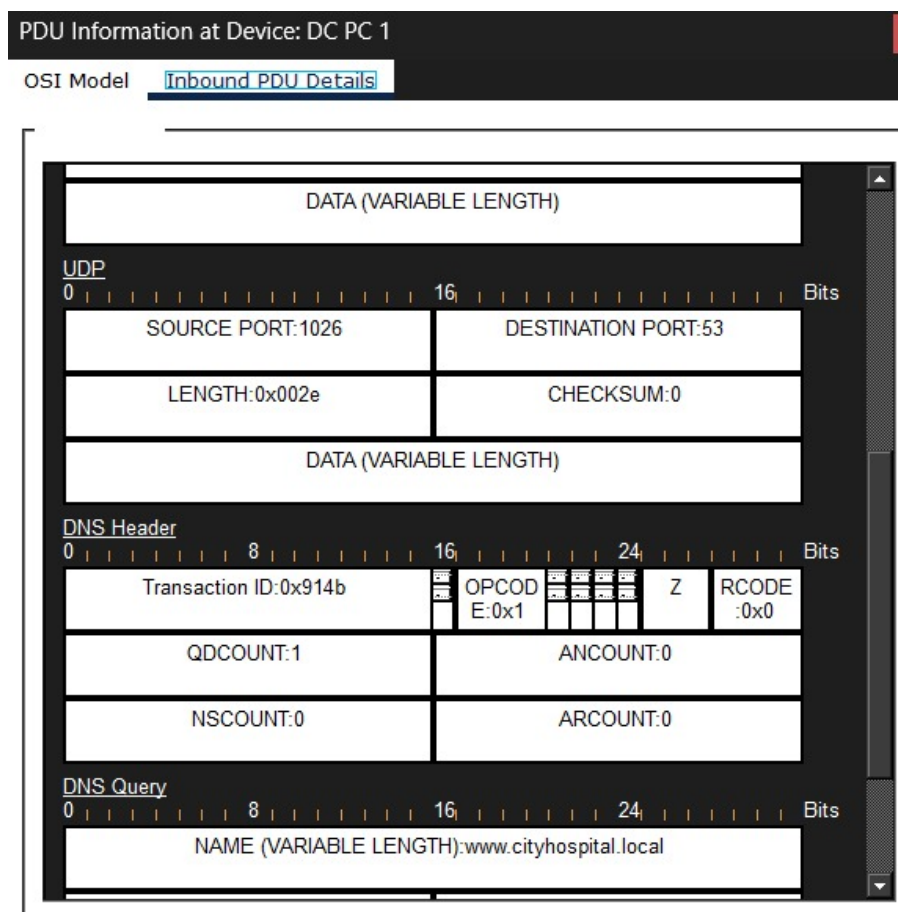


Figure 6: DNS PDU details showing UDP destination port 53 and the query for `www.cityhospital.local`.

6.2 HTTP Request and Response Analysis

After DNS resolves the website name to the web server IP address, the client begins HTTP communication with the web server. The HTTP PDU details show TCP being used, with source port 80 in the shown response. TCP is appropriate for HTTP because web page data must arrive correctly, in order, and without missing segments. TCP provides reliability through acknowledgments, sequencing, and retransmission when needed.

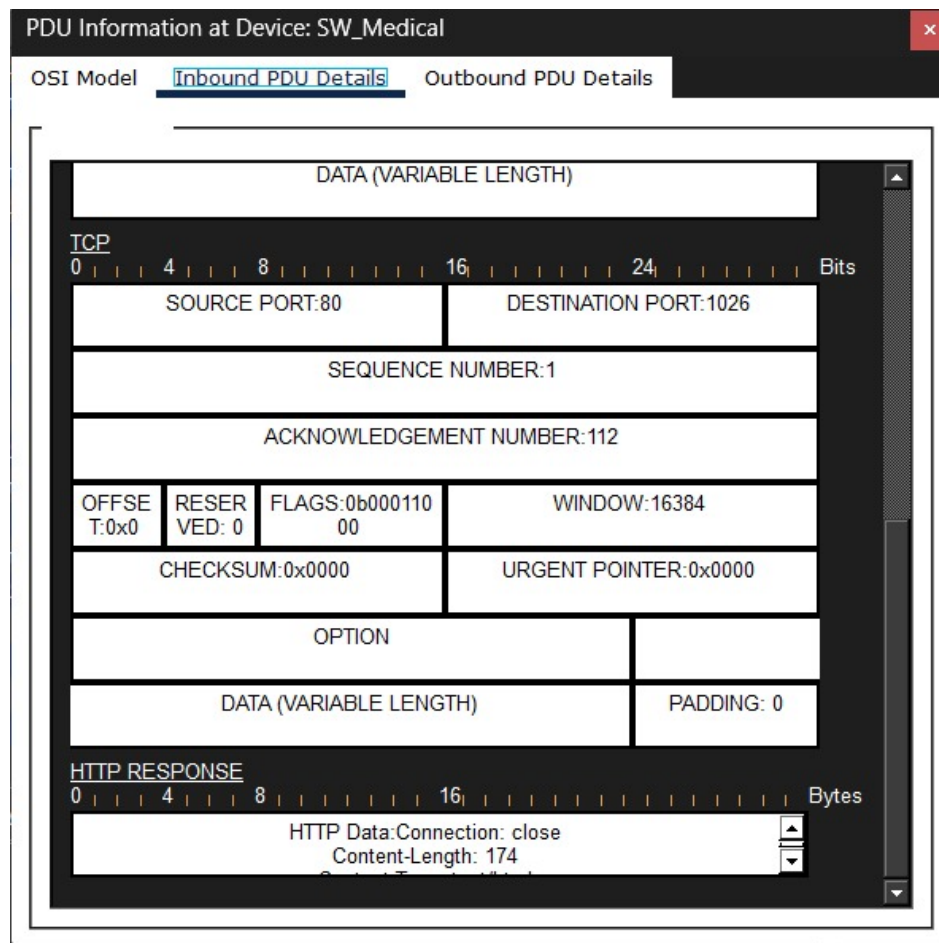


Figure 7: HTTP PDU details showing TCP transport information for the intranet web response.

6.3 DNS vs. HTTP Comparison

Protocol	Transport Protocol	Port	Reason
DNS	UDP	53	DNS uses fast, lightweight request-response communication. Most DNS queries are small, so UDP reduces overhead.
HTTP	TCP	80	HTTP needs reliable delivery so that web-page content arrives complete, ordered, and error-free.

Table 5: Comparison between DNS and HTTP transport-layer behavior.

Conclusion

The City Smart Hospital Network Design successfully implements a scalable and redundant hospital network using VLANs, FLSM subnetting, DHCP, OSPF routing, NAT

overload, DNS, and web services. The network separates departments into independent broadcast domains, supports internal and external connectivity, and includes a Remote Radiology Clinic expansion. The failover test demonstrated that OSPF can reconverge after the primary Medical-to-Core link fails, allowing traffic to continue through the backup Medical-to-Admin path. Finally, the protocol analysis confirmed the difference between DNS over UDP and HTTP over TCP during access to the hospital intranet website.